COS 214 - Practical 6

Git Push and Pray

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Task 1: Research Workflow Management Systems

A workflow is a structured sequence of tasks that are carried out in a particular order to achieve an outcome. A Workflow Management System manages this process by coordinating tasks, considering their dependencies and directing work to the appropriate people or system [8]. From looking at different workflow systems, it became clear that a workflow management system is more than just a list of tasks. It needs to manage how work is created, assigned, routed, completed and monitored throughout the process.

One of the most important concepts identified during the research was the difference between a workflow definition and a workflow instance. A workflow definition is the reusable blueprint of a process. It describes the tasks, actions, transitions and overall structure of the workflow. A workflow instance is one actual execution of that definition and contains its own runtime information, such as its current state and progress [7]. This distinction is important for TIVIDY because the same workflow definition should be able to create multiple independent instances at the same time.

The research also showed that certain responsibilities appear repeatedly across different workflow systems. These include managing workflow, definitions and instances, creating and assigning work items, routing work between activities, responding to events and dependencies, recording workflow history, monitoring execution and communicating with external systems. Workflow systems can also contain nested subprocesses, where one activity contains several smaller activities, gateways and events [4].

Another important finding was that some parts of the system are expected to change independently. For example, the way work is assigned may depend on roles, workload or other rules. Approval and escalation processes may also differ between organisations. Decisions such as routing, assignment and approval can be kept as configurable rules instead of being hard coded into the workflow engine [1]. This suggests that our system should keep these changing behaviours separate from its core workflow execution.

External systems are another area that needs to remain flexible. Workflow systems can communicate with other systems using connectors, API's and messaging [1]. For TIVIDY, this means that the workflow manager should coordinate with external systems without becoming responsible for the functionality of those systems. The research also showed that workflow definitions may change while existing workflows are still running. AWS Simple Workflow Service handles this using versioning, allowing existing instances to continue with their original definition while new instances use the newer version [5].

Overall, the research helped identify what TIVIDY actually needs to be designed around. TIVIDY should be a generic and reusable workflow manager, rather than a system containing logic for one specific process. It needs to manage workflow execution, work items, routing, participants, state, events, history and external integrations. At the same time, behaviours such as routing strategies, approval rules, notifications and external system interfaces should be flexible enough to change without changing the core workflow engine.

References

[1] Connector types. (n.d.). Camunda 8 Docs. Retrieved October 2, 2026, from https://docs.camunda.io/docs/components/connectors/connector-types/

[2] Defining a workflow action. (n.d.). IBM. Retrieved October 2, 2026, from https://www.ibm.com/docs/en/openpages/9.2.x?topic=actions-defining-workflow-action

[3] Designing a workflow. (n.d.). IBM. Retrieved October 2, 2026, from https://www.ibm.com/docs/en/openpages/9.2.x?topic=workflows-designing-workflow

[4] Subprocess. (n.d.). Flowable Enterprise Documentation. Retrieved October 2, 2026, from https://documentation.flowable.com/latest/reactmodel/bpmn/reference/subprocess

[5] Versioning - Amazon Simple Workflow Service. (n.d.). AWS Documentation. Retrieved October 2, 2026, from https://docs.aws.amazon.com/amazonswf/latest/developerguide/swf-dev-adv-versioning. html

[6] What Is a Workflow? (n.d.). IBM. Retrieved October 2, 2026, from https://www.ibm.com/think/topics/workflow

[7] Workflow definitions and workflow instances. (n.d.). IBM. Retrieved October 1, 2026, from https://www.ibm.com/docs/en/openpages/9.2.x?topic=fundamentals-workflow-definitionsworkflow-instances

[8] Workflow Management System - Glossary | CSRC. (n.d.). NIST Computer Security Resource Center. Retrieved October 2, 2026, from https://csrc.nist.gov/glossary/term/workflow_management_system

Task 2: Define the Scenario and System

1. Initial Scope of TIVIDY
   TIVIDY is a generic workflow management system that coordinates workflows Its main responsibilities are to:
   - Store and manage workflow definitions and versions
   - Create and manage workflow instances
   - Create, assign and track work items
   - Route work using configurable rules
   - Mange participants, approvals, reactions and escalation
   - Handle states, events and dependencies
   - Support nested activities (sub processes)
   - Communicate with external systems
   - Record history and audit information
   - Monitor running workflows

2. System Boundaries
   TIVIDY is responsible for coordinating and tracking work, but not for performing the specialised business work itself.
   For example: TIVIDY can assign a doctor an assessment task and track its completion. It also does not replace the hospital's patient, laboratory, pharmacy or billing systems.

3. Workflow definition vs workflow instance
   Workflow definition: reusable blue print of a process. It contains activities, transitions, routing rules, dependencies, approval and escalation rules, events, subprocesses and its version.
   A workflow instance: represents one actual execution of a definition. It contains runtime information such as its current activity and state, assigned participants, completed or pending work, events, errors and history.
   This means that multiple independent instances can run from the same definition.

4. Parts expected to vary
   The following parts should be able to change independently:
   - Assignment and routing: eg. role based, workload based or rule based.
   - Approval and escalation rules: different processes may require different approval levels.
   - Work item behaviour: validation, priority, notifications and security.
   - External integrations: different systems may require different interfaces.
   - Reporting: organisations may require different workflow statistics and reports.

5. Workflow scenario: Hospital Patient Admissions and Treatment

   Organisation/Environment
   The scenario takes place in a hospital where different departments and roles must coordinate a patient's admission, assessment, treatment and discharge.

   Process
   The workflow manages a patient journey:
   Patient arrives -> Register patient -> Nurse assessment -> Doctor assessment -> determine treatment -> treatment approval (if required) -> treatment -> monitoring -> discharge

Participants/Roles

- Admission staff: register the patient and capture information
- Nurse: perform the initial assessment and record observations
- Doctor: assess the patient and determine treatment
- Senior Doctor/Specialist: handle escalated cases and higher risk approvals
- Laboratory staff: perform requested tests and return results
- Pharmacist: review and dispense approved medication
- Discharge staff: complete the discharge process

Major Work

1. Register patient
2. Perform initial assessment
3. Doctor assessment
4. Request and review tests
5. Determine treatment
6. Obtain approval if required
7. Perform treatment
8. Monitor patient
9. Assess readiness for discharge
10. Discharge patient

Decisions and Routing

- Emergency: follow emergency route
- Tests required: Send request to laboratory
- Treatment requires approval: Send for approval
- Approved?: Continue treatment otherwise escalate/reassess
- Ready for discharge?: Discharge otherwise continue monitoring

Approvals and Dependencies

Some treatments require approval from a doctor or senior doctor.
Activities can also depend on previous activities
Eg. Doctor assessment -> test request -> laboratory test -> result -> doctor -> review The doctor cannot review the result until the laboratory test has been completed

Failures/Exceptions

- Lab failure: retry or alternative route
- Approval timeout: escalate to a senior doctor
- Treatment rejected: reassess treatment
- Patient condition worsens: trigger an emergency workflow
- External system unavailable: retry, wait, notify or escalate

External Systems

- Patient information system
- Laboratory system
- Pharmacy system
- Notification system

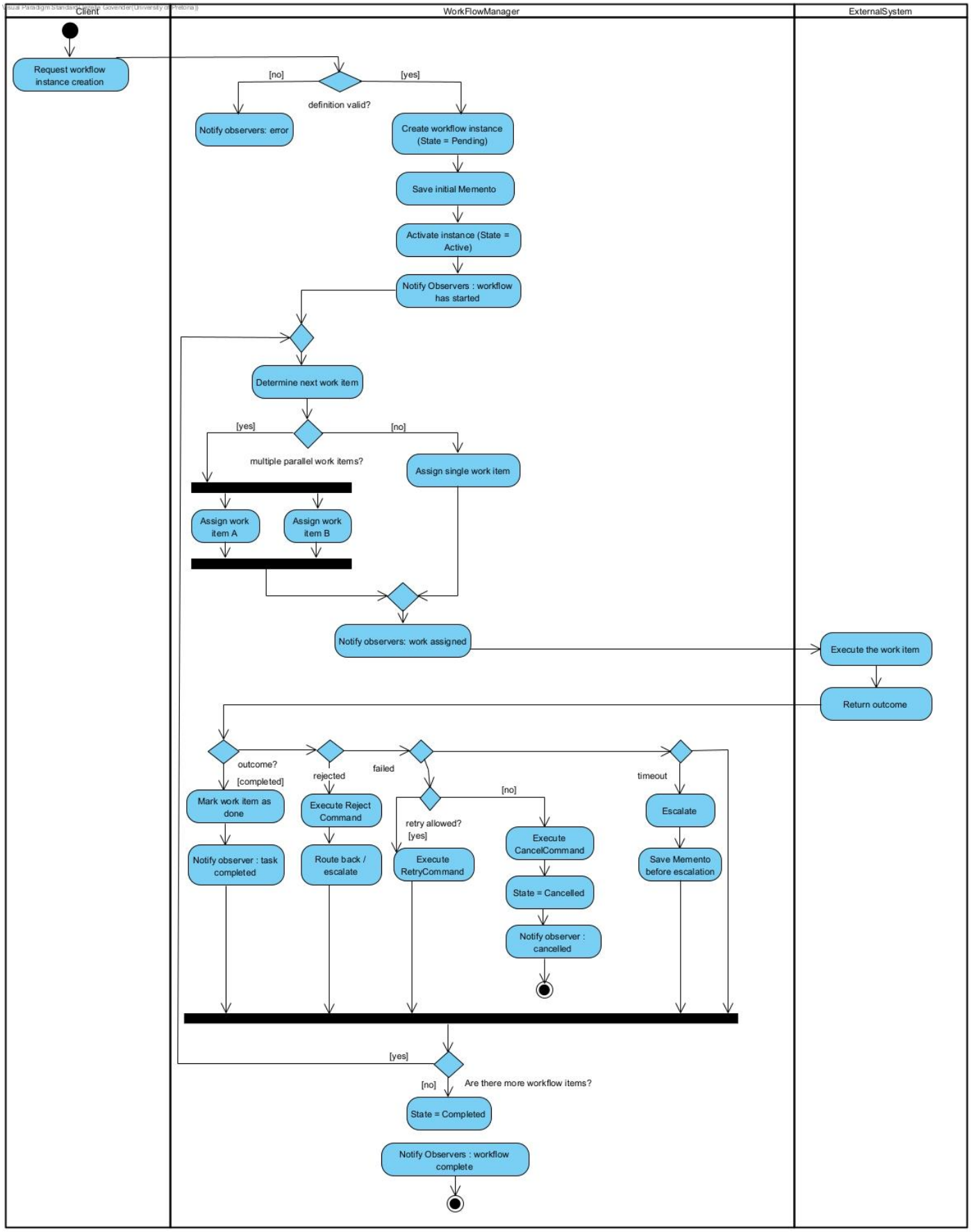
TIVIDY coordinates these interactions but does not replace the external systems.

Expected Outcome

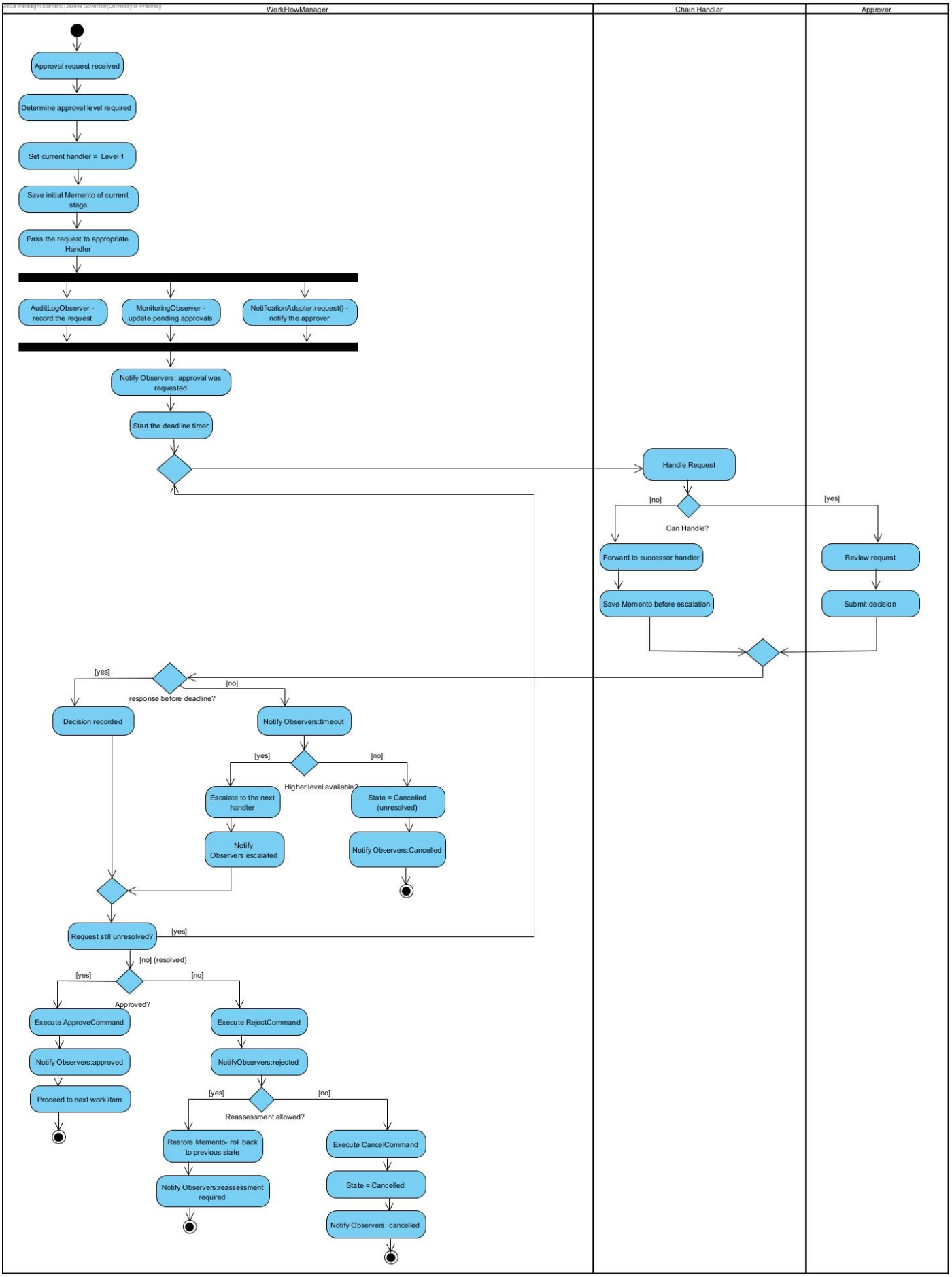
The workflow should result in successful treatment and discharge, continued treatment, escalation or controlled failure/cancellation, while retaining the workflow history.

Task 3: UML Activity Diagrams

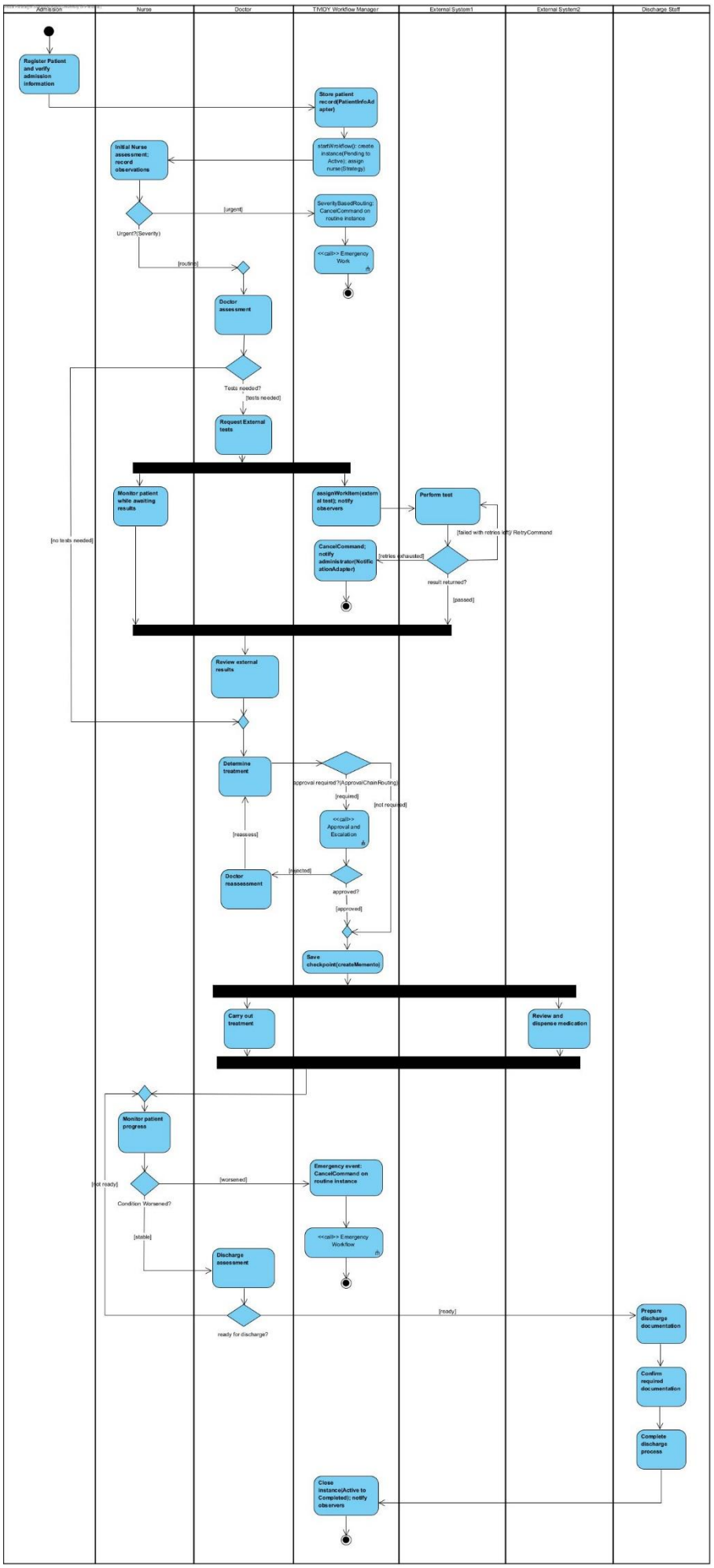
1. A generic workflow execution showing the progression of a workflow instance from creation or start until completion or another appropriate final outcome



2. Another important generic TIVIDY workflow, such as work assignment and completion, approval and escalation, failure and recovery, or event handling.



3. An important workflow from your selected scenario.



Task 4: Design Patterns

WorkflowManager

The WorkflowManager is the central coordinator of TIVIDY. Its responsibility is to manage and coordinate running WorkflowInstance objects rather than performing specialised business work itself. A workflow definition describes how a process may behave, while each WorkflowInstance represents an independent execution with its own state, progress, participants, data and history. The manager therefore determines which actions, strategies, integrations or other workflow mechanisms need to be invoked as an instance progresses through its lifecycle.

Workflow Execution and Lifecycle

The execution and lifecycle subsystem is centred around WorkflowInstance and combines the Command, State and Memento patterns. The State pattern represents the current lifecycle state of a workflow instance, such as Pending, Active, Completed or Cancelled, and allows behaviour to change according to the current state. The Command pattern represents workflow actions such as approving, rejecting, retrying or cancelling an operation. The WorkflowManager acts as the Invoker and invokes commands on the appropriate WorkflowInstance, which acts as the Receiver. The Memento pattern allows the runtime state of a WorkflowInstance to be captured and restored when rollback or recovery is required. The WorkflowInstance therefore acts as the Originator, while the WorkflowManager acts as the Caretaker by storing Memento objects.

Assignment and Routing

The assignment and routing subsystem determines where work should go and who should perform it. The Strategy pattern allows the WorkflowManager to select different assignment and routing algorithms without changing the core workflow engine. Assignment strategies may include role-based, workload-based and rule-based assignment, while routing strategies may include severity-based, approval-chain and event-based routing. The Composite pattern supports the hierarchical organisation of participants, allowing employees and supervisors to be represented uniformly. The Iterator pattern can be used to traverse participants, stages and other workflow collections without exposing their internal storage structure.

Events and Integration

The events and integration subsystem allows TIVIDY to react to events occurring during workflow execution and to communicate with external systems. The Observer pattern allows the WorkflowManager to notify multiple interested observers when a workflow event occurs. These observers may perform tasks such as recording audit information, updating monitoring information, responding to task completion, triggering additional workflow behaviour or requesting notifications. The Adapter pattern separates TIVIDY from the interfaces of external systems. TIVIDY communicates through the common external-system abstraction while concrete adapters translate requests for systems such as the participant-information and notification systems. This allows external integrations to change independently of the workflow engine.

Escalation

The escalation subsystem uses the Chain of Responsibility pattern to handle requests that may need to move through multiple levels of responsibility. A request can begin with a Level 1 handler and be passed to Level 2 or Level 3 when the current handler cannot resolve it or when escalation is required. The WorkflowManager initiates or coordinates the escalation process. This is particularly useful when a workflow reaches a condition such as an approval timeout, unresolved issue or other situation requiring progressively higher levels of responsibility.

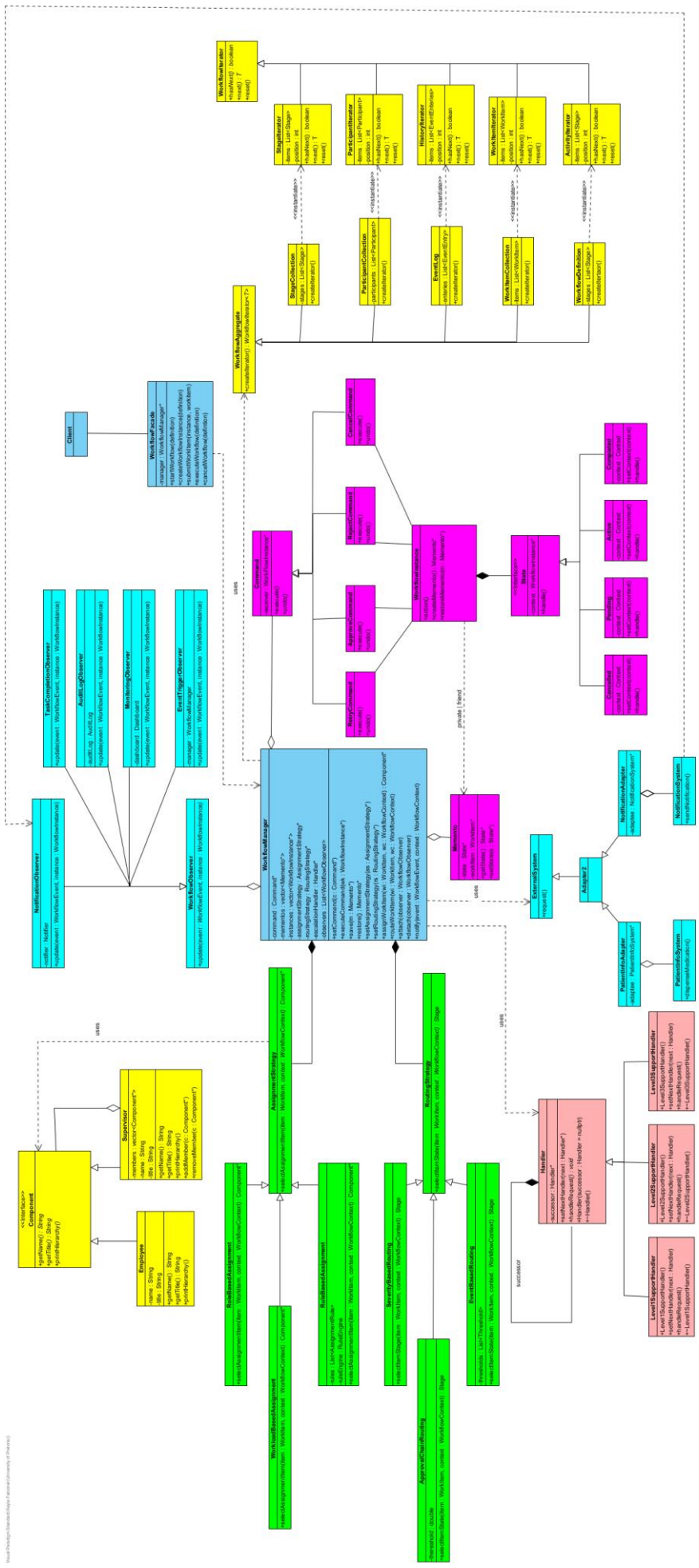
Workflow Structure and Traversal

The workflow structure subsystem represents the organisational structure and collections that make up a workflow. The Composite pattern allows hierarchical structures such as supervisors containing employees to be treated through a common component interface. The Iterator pattern provides a standard way of traversing workflow definitions, stages, work items, participants and event logs without exposing how those collections are internally stored. These patterns allow the workflow engine to operate on complex structures while keeping the traversal and hierarchy mechanisms separate from workflow execution logic.

Overall Lifecycle

During execution, the WorkflowManager coordinates the lifecycle of each WorkflowInstance. An instance progresses through its State-controlled lifecycle while the manager invokes Commands when actions are required. Assignment and routing Strategies determine where work should move and who should receive it. Events generated during execution are distributed through the Observer mechanism, which allows monitoring, auditing, notifications and other reactions to occur independently. When escalation is required, the Chain of Responsibility determines which level should handle the request. When communication with an external system is required, an Adapter translates the TIVIDY request into the interface expected by that system. Mementos allow the manager to retain previous workflow states so that an instance can be restored when appropriate. Throughout the process, Composite and Iterator provide mechanisms for managing and traversing the workflow's participants and structural collections. The WorkflowManager therefore acts as the central orchestrator while the individual design patterns provide specialised, interchangeable mechanisms for managing different aspects of a WorkflowInstance's lifecycle.

Task 5: UML Class Diagram



Key:

 - Workflow Execution and Lifecycle

 - Assignment and Routing

 - Events and Integration

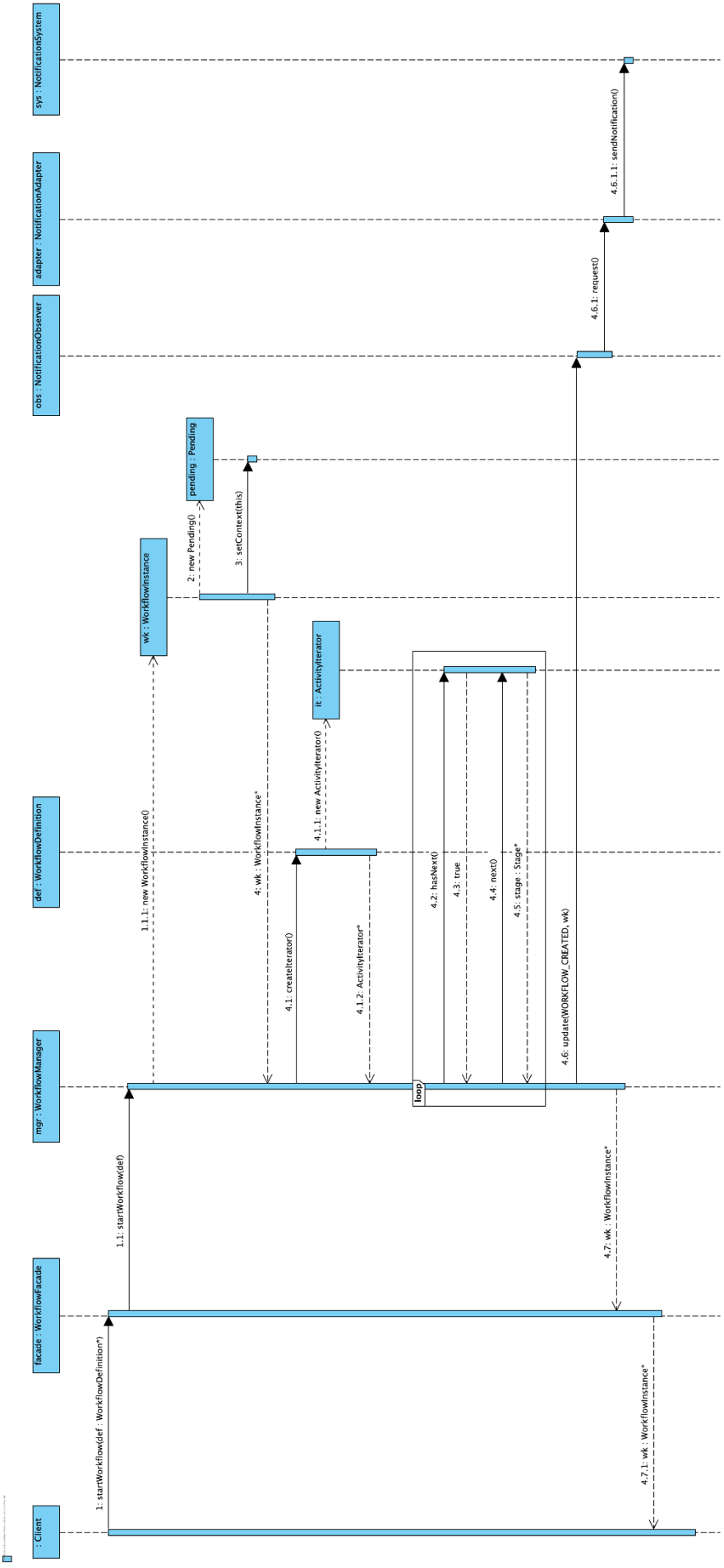
 - Escalation

 - Assignment and Routing, Workflow Structure and Traversal

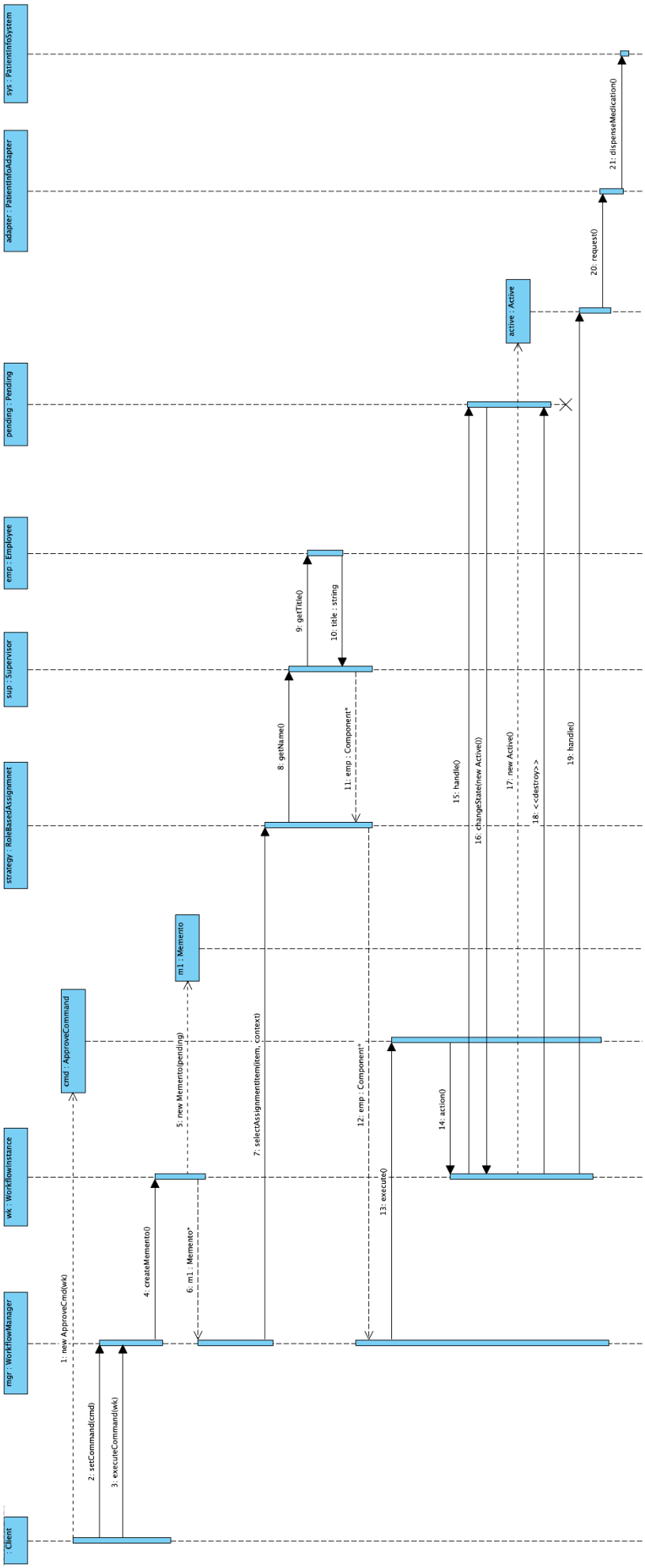
 - Workflow Execution and Lifecycle, Assignment and Routing, Events and Integration, Escalation

Task 6: Runtime Behaviour Diagrams

UML Sequence Diagram



UML Sequence Diagram



UML State Diagram

